

MATH0119 Dynamical Systems

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0119
<i>Level:</i>	7 (UG), 7 (PG)
<i>Normal student group(s):</i>	UG: Y3, Y4 Mathematics programmes PG: MSc Mathematics
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	90% examination, 10% coursework
<i>Normal Pre-requisites:</i>	MATH0017 (Measure Theory) MATH0019 (Multivariable Analysis) or MATH0020 (Curves and Surfaces)
<i>Lecturer:</i>	Dr S Ghazouani

Course Description and Objectives

In this course, we will give an introduction to the theory of dynamical systems. A dynamical system is given by a transformation of a topological space which we think of as the law driving the evolution of a system over time (such as the evolution over time of the solar system, a population, the weather, you-name-it). Our goal will be to develop the tools and the language which allow for the general treatment of such problems: this will involve a good dose of topology, analysis and measure-theory.

Recommended Texts

- Katok and Hasselblatt, *Introduction to the Modern Theory of Dynamical Systems*
- Brin and Stuck, *Introduction to Dynamical Systems*
- Einsiedler and Ward, *Ergodic theory: with a view towards number theory*

Detailed Syllabus

- Quasi-periodic dynamics. Linear flows on the torus, circle homeomorphisms, rotation number, Poincaré theorem.
- Chaotic dynamics. Doubling map, shift maps, Smale’s horseshoe
- Hyperbolic dynamics. Hyperbolic fixed points in $\mathbb{R}^2$, Stable Curve Theorem, Local Stability, homoclinic intersections
- Ergodic theory. Invariant measures, Ergodicity, Birkhoff’s Ergodic Theorem